Anushree C

co23btech11003@iith.ac.in | aanuushree@gmail.com | +91 8939596811 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.51
St. Johns Public School	Class XII (CBSE)	2023	95.40%
SRDF. Vivekananda Vidyalaya	Class X (CBSE)	2021	90.20%

INTERNSHIP

Microsoft | SWE Intern

- **Azure SRE Agent Architecture:** Architected an autonomous SRE agent for the ATS team using **ICM**, **Kusto log**, and **multi-agent routing** for automated incident triage and root-cause analysis.
- **Workflow Optimization & Feedback:** Reduced manual triage from **2-3 hours to minutes** via a post-incident DRI feedback loop, while automating daily scheduled tasks for **pipeline health analytics**.
- **Policy Builder Automation:** Built end-to-end policy generation, validation, and ADO work-item creation for AdQuality, converting a **1-2 hour manual process into a single-click workflow in seconds**.

Research Intern under Dr. Sai Sidhardh (IIT - Hyderabad)

- Implemented **PINNs** to solve fractional-order viscoelasticity equations via an **optimized Caputo derivative scheme**. Initially validated the framework on single-degree-of-freedom systems, achieving strong agreement with analytical solution, and later extended it to a simply supported viscoelastic beam.

PROJECTS

Voice Activity Recognition

- Built a voice activity detection system that classifies speech as voiced, unvoiced, or silent using autocorrelation, and validated it against manually labeled speech data.
- Tested it on a standard Keele speech dataset across 10 speakers, reaching a 25% false-alarm rate for detecting voiced speech, and under 4% error in pitch estimation.

Autograd Engine:

- Engineered a NumPy based automatic differentiation engine from scratch, dynamic computation graphs with backpropagation through arithmetic and matrix operations, mirroring PyTorch's core design.
- Built a Linear layer, ReLU/Sigmoid activations, a Sequential container, and Cross-Entropy loss on top, with an SGD optimizer and a graph visualizer verifying gradient flow end-to-end.

Interval Asynchronous Parallel SGD

- Built a distributed Interval-Asynchronous SGD optimization system using ZeroMQ to execute non-blocking asynchronous message passing between worker nodes and a central dispatcher.
- Implemented lock-free interval accept/reject logic to enforce bounded gradient staleness, achieving model convergence within 0.5% of ground truth on regression tasks.

RISCV Simulator

- Built an Assembler which converts assembly code into machine code.
- Extended it into a full simulator supporting memory management, cache simulation, and register inspection, wrapped in a user-friendly interface.

SKILLS

- **Programming:** Python, C, C++, JavaScript, TypeScript, Assembly [RISC-V], MATLAB, HTML/CSS
- **Frameworks & Libraries:** ReactJS, NextJS, NodeJS, Flask, REST APIs, OpenMPI

- **Data Science & ML:** PyTorch, NumPy, Pandas, Matplotlib, Seaborn
- **Databases & DevOps:** Docker, MySQL, MongoDB, Azure DevOps, Git, Bash, Linux
- **Core Concepts:** DBMS, Object-Oriented Programming

RELEVANT COURSES

- **Computer Science:** Data Structures & Applications, Computer Architecture, Operating Systems, Distributed Computing
- **AI and Scientific Computing:** Fundamentals of Machine Learning**,** Fundamentals of Scientific Computing, Computer Aided Numerical Methods I & II, Machine Learning and Data Analytics in Materials, High Performance Computing
- **Electrical:** Digital Circuits, Introduction to HDL, Speech Systems
- **Mathematics:** Probability, Linear Algebra, Transform Techniques, Differential Equations, Calculus I & II, Statistics

POSITION OF RESPONSIBILITY

Lambda Senior Core - Software Development Club of IITH

- Built and deployed the Key Management software for IITH, currently used campus-wide; developed a themed gaming club website using Next.js and conducted technical sessions for juniors.

Sunshine Mentor

- Core organizer of the inaugural National Well-Being Conclave at IIT Hyderabad with the Ministry of Education; mentored 30+ juniors through monthly academic and well-being sessions.

Class Representative

- Led the launch of the first official Computational Engineering department website; liaised with faculty/administration to resolve academic concerns and organized Department Day with faculty, students, and external speakers.

EXTRACURRICULAR

- Carnatic vocalist with 10+ years of performance experience; winner of a televised Bhajan competition on DD TV and participant in Bhajan Samrat. Keyboard player with extensive classical music performance experience.
- Built a Food Serving Robot, reflecting interest in robotics.